Township Infrastructure Management System (TIMS)

Complete Actor Roles, Functionalities, Modules & End-to-End Workflow

Updated Operational Specification | October 2026

Project Scope

TIMS is a township-level infrastructure management platform that manages the complete lifecycle of infrastructure issues: reporting, validation, triage, contractor estimation, AMC-based approval, execution, billing, payment authorization, and resident verification.

The system uses Annual Maintenance Contracts (AMCs) with predefined contractor rates. TIMS therefore manages contractual rates, work quantities, approvals and billing rather than township-owned inventory.

1. System Actors

Actor	Type	Primary Responsibility
RWA Representative	External	Report infrastructure issues, track progress, and verify completed work.
Field Contractor	External	Inspect issues, prepare AMC-based estimates, execute repairs, and submit completion evidence.
Desk Clerk	Internal	Validate complaints, perform triage, create work orders, assign contractors, and monitor queues.
Department Head	Internal	Supervise department operations, handle escalations, and review and approve every contractor estimate.
Finance Clerk	Internal	Verify estimates and invoices against AMC rates, verify RWA verification success, and authorize payment.
Township COO	Internal	Monitor township-wide KPIs, expenditure, and department performance.
System Administrator	Internal	Manage users, roles, departments, township configuration, contractors, AMCs, and system settings.

2. RWA Representative

The RWA Representative connects residents with township infrastructure operations.

- **Report an Issue:** Create a complaint with category, description, location, severity, and photographs across Civil, Electrical, or Water.
- **Track Complaints:** View ticket status, assigned contractor, expected completion, and progress.
- **View Resolution Evidence:** Inspect before/after photographs and completion information uploaded by the contractor.
- **Verify Completed Work:** Confirm that the issue has been resolved on site.

3. Field Contractor

The Field Contractor performs technical inspection and ground-level execution under the applicable AMC.

- **Receive Work Orders:** View assigned work orders in the contractor queue.
- **Inspect the Site:** Visit the location, inspect the issue, and record required work quantities and labour.
- **Prepare AMC-Based Estimate:** Select contractual rate items from the AMC rate card; TIMS calculates the total automatically.
- **Execute Work:** Transition work to In Progress, perform repairs, record actual quantities, and upload mandatory completion evidence.

4. Desk Clerk

Desk Clerks are operational users responsible for departmental triage and work-order coordination.

- **Complaint Validation:** Review incoming complaints, photographs, location, and department category.
- **Work Order Creation:** Convert validated complaints directly into formal work orders and assign to empanelled AMC contractors.
- **Queue Monitoring & Escalation:** Monitor pending, assigned, in-progress, and completed queues; escalate exceptions to the Department Head.

5. Department Head

The Department Head manages an infrastructure department and serves as the managerial approval layer.

- **Department Monitoring:** Monitor open work orders, pending approvals, completions, and expenditure.
- **Escalation Handling:** Resolve field exceptions and contractor delays.
- **Mandatory Estimate Approval:** Review and approve or reject every contractor estimate regardless of amount. Return unapproved estimates for revision.

6. Finance Clerk

The Finance Clerk performs financial audits and authorizes payments upon verified work completion.

- **Estimate and Invoice Verification:** Compare approved estimates, actual quantities, and invoices against AMC rates.
- **Variance Handling:** Flag discrepancies between estimated and billed figures.
- **Payment Authorization Gating:** Checks: *'Has RWA verification succeeded?'* If no, Finance waits; if yes, Finance authorizes payment.

7. Township COO

- **Township Dashboard:** View complaints, open work orders, departmental performance, and expenditure.
- **Performance & Strategic Oversight:** Review contractor performance and multi-department service delivery.

8. System Administrator

- **User & Role Management:** Administer accounts, permissions, and department mappings.
- **Township & AMC Configuration:** Maintain sectors, blocks, contractor profiles, validity dates, and AMC rate cards.
- **Audit & Security:** Track system audit logs and critical administrative changes.

9. Township & Location Hierarchy

Township → Sector / Zone → Block / Street → Infrastructure Asset → Complaint → Work Order

10. Infrastructure / Asset Management

TIMS associates complaints with physical assets: Civil (roads, drains, footpaths), Electrical (streetlights, transformers, panels), and Water (pipelines, pumps, valves).

11. Annual Maintenance Contract (AMC) Management

Predefined contractual rate cards ensure consistent pricing across repairs:

Item	Quantity	Contract Rate	Amount
Electrician	2 hours	₹500 / hour	₹1,000
General Labour	3 hours	₹300 / hour	₹900
Cable Work	10 m	₹150 / m	₹1,500
Total			**₹3,400**

12. Complete End-to-End Workflow

- **1. Reporting:** RWA Representative submits an infrastructure complaint with category, description, location, severity, and photos.
- **2. Desk Triage:** Desk Clerk reviews complaint details and validates departmental assignment.
- **3. Work Order Creation:** Desk Clerk creates a formal work order and assigns an empanelled AMC contractor.
- **4. Contractor Inspection:** Contractor inspects the site and logs required work items and quantities.
- **5. AMC-Based Estimation:** Contractor submits an estimate using AMC rate card items; system computes the total cost.
- **6. Department Head Approval:** Department Head reviews the estimate. Approval is mandatory for every estimate. If revision is needed, estimate returns to the contractor to update and resubmit.
- **7. Work Execution:** Contractor proceeds with the work, records actual quantities, and performs repairs.
- **8. Completion Evidence:** Contractor uploads completion details and evidence photos, marking the work Completed.
- **9. RWA Verification:** RWA Representative conducts physical verification and records completion.
- **10. Finance Audit:** Finance receives the invoice and checks it against AMC rates and approved estimates.
- **11. Payment Gate & Authorization:** Finance verifies: 'Has RWA verification succeeded?' If No, Finance waits; if Yes, Finance authorizes payment.
- **12. Disbursement & Closure:** Payment is disbursed and recorded, and the work order lifecycle closes.

13. Estimate Approval Policy

Estimated Amount	Approval Route
All Estimates (Mandatory)	Department Head approval required for every estimate, regardless of its amount.

14. Work Order Status Lifecycle

REPORTED → UNDER_REVIEW → WORK_ORDER_CREATED → ASSIGNED → INSPECTION → ESTIMATE_SUBMITTED → PENDING_APPROVAL → APPROVED → IN_PROGRESS → COMPLETED → PENDING_VERIFICATION → VERIFIED → PAYMENT_AUTHORIZED → CLOSED

Revision Loop: PENDING_APPROVAL → REVISION_NEEDED → ESTIMATE_SUBMITTED → PENDING_APPROVAL

15. Core TIMS Modules

1. User & RBAC | 2. Township & Location | 3. Asset Management | 4. Complaint Management | 5. Work Order Management | 6. Contractor Management | 7. AMC Management | 8. Departmental Estimate Approvals | 9. RWA Verification | 10. Finance Audit & Payment | 11. Notifications | 12. Analytics | 13. Audit Logs

16. Scope Boundary

Included: Infrastructure ticket lifecycle, AMC pricing, work orders, estimate approvals, execution, evidence, RWA verification, finance audit, gated payment authorization, and analytics.

Excluded: Township-owned physical warehouse and stock inventory management.

17. Overall System Chain

Township → Sector → Block → Infrastructure Asset → Complaint → Work Order → AMC Estimate → Department Head Approval → Execution → RWA Verification → Finance Audit → RWA Verification Check (Gate) → Payment Authorization → Closure